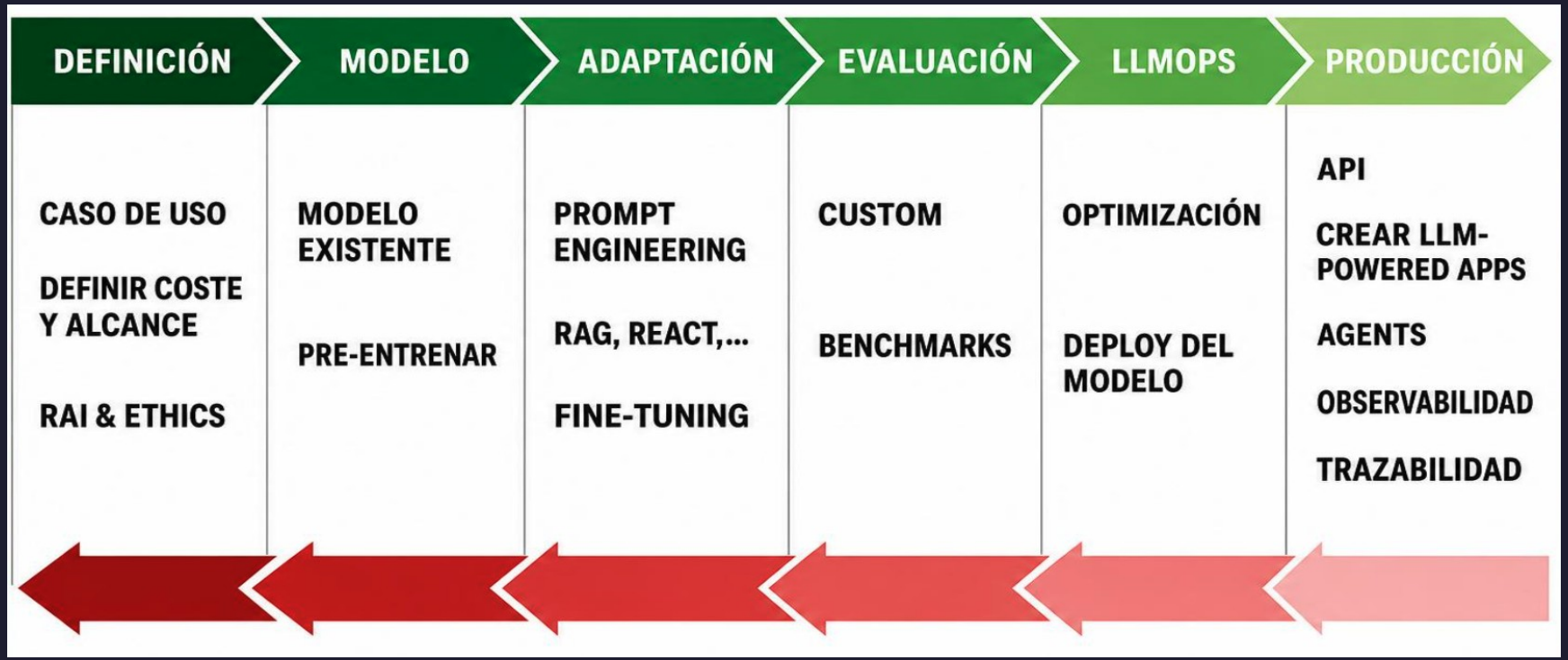
keep coding

AI ENGINEERING

Fine-tuning


DEFINICIÓN
MODELO
ADAPTACIÓN
EVALUACIÓN
LLMOPS
PRODUCCIÓN
CASO DE USO
DEFINIR COSTE Y ALCANCE
RAI & ETHICS
MODELO EXISTENTE
PRE-ENTRENAR
PROMPT ENGINEERING
RAG, REACT,...
FINE-TUNING
CUSTOM
BENCHMARKS
OPTIMIZACIÓN
DEPLOY DEL MODELO
API
CREAR LLM-POWERED APPS
AGENTS
OBSERVABILIDAD
TRAZABILIDAD

Pre-entrenamiento vs fine-tuning

keep coding

- Pre-entrenamiento · el modelo crudo: potente pero caótico
- SFT · le da forma útil sin reentrenarlo entero
- La diferencia entre el monstruo y algo con lo que hablar



Pre-entrenar cuesta una fortuna · fine-tunear no

	Pre-entrenar	Fine-tunear
Datos	10 TB de internet	100k ejemplos curados
Hardware	6.000 GPU	1 GPU
Coste	~2.000.000 $	decenas de $
Tiempo	12 días	1 día
Resultado	un loro estocástico	un modelo que cumple

De GPT a GPT-3 · escalar funciona

keep coding

Cada salto de escala trae capacidades nuevas inesperadas



Improving Language Understanding by Generative Pre-Training

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Abstract

Natural language understanding comprises a wide range of diverse tasks such as textual entailment, question answering, semantic similarity assessment, and document classification. Although large unlabeled text corpora are abundant, labeled data for learning these specific tasks is scarce, making it challenging for discriminatively trained models to perform adequately. We demonstrate that large gains on these tasks can be realized by *generative pre-training* of a language model on a diverse corpus of unlabeled text, followed by *discriminative fine-tuning* on each specific task. In contrast to previous approaches, we make use of task-aware input transformations during fine-tuning to achieve effective transfer while requiring minimal changes to the model architecture. We demonstrate the effectiveness of our approach on a wide range of benchmarks for natural language understanding. Our general task-agnostic model outperforms discriminatively trained models that use architectures specifically crafted for each task, significantly improving upon the state of the art in 9 out of the 12 tasks studied. For instance, we achieve absolute improvements of 8.9% on commonsense reasoning (Stories Cloze Test), 5.7% on question answering (RACE), and 1.5% on textual entailment (MultiNLI).

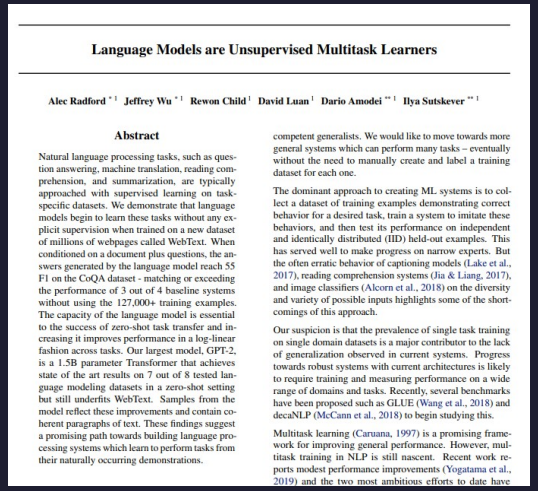
1 Introduction

The ability to learn effectively from raw text is crucial to alleviating the dependence on supervised learning in natural language processing (NLP). Most deep learning methods require substantial amounts of manually labeled data, which restricts their applicability in many domains that suffer from a dearth of annotated resources [61]. In these situations, models that can leverage linguistic information from unlabeled data provide a valuable alternative to gathering more annotation, which can be time-consuming and expensive. Further, even in cases where considerable supervision is available, learning good representations in an unsupervised fashion can provide a significant performance boost. The most compelling evidence for this so far has been the extensive use of pre-trained word embeddings [10, 39, 42] to improve performance on a range of NLP tasks [8, 11, 26, 45].

Leveraging more than word-level information from unlabeled text, however, is challenging for two main reasons. First, it is unclear what type of optimization objectives are most effective at learning

2018 · GPT · 117M

decoder-only · siguiente token



Language Models are Unsupervised Multitask Learners

Alec Radford [*1] Jeffrey Wu [*1] Rewon Child [1] David Luan [1] Dario Amodei [**1] Ilya Sutskever [**1]

Abstract

Natural language processing tasks, such as question answering, machine translation, reading comprehension, and summarization, are typically approached with supervised learning on task-specific datasets. We demonstrate that language models begin to learn these tasks without any explicit supervision when trained on a new dataset of millions of webpages called WebText. When conditioned on a document plus questions, the answers generated by the language model reach 55 F1 on the CoQA dataset - matching or exceeding the performance of 3 out of 4 baseline systems without using the 127,000+ training examples. The capacity of the language model is essential to the success of zero-shot task transfer and increasing it improves performance in a log-linear fashion across tasks. Our largest model, GPT-2, is a 1.5B parameter Transformer that achieves state of the art results on 7 out of 8 tested language modeling datasets in a zero-shot setting but still underfits WebText. Samples from the model reflect these improvements and contain coherent paragraphs of text. These findings suggest a promising path towards building language processing systems which learn to perform tasks from their naturally occurring demonstrations.

competent generalists. We would like to move towards more general systems which can perform many tasks – eventually without the need to manually create and label a training dataset for each one.

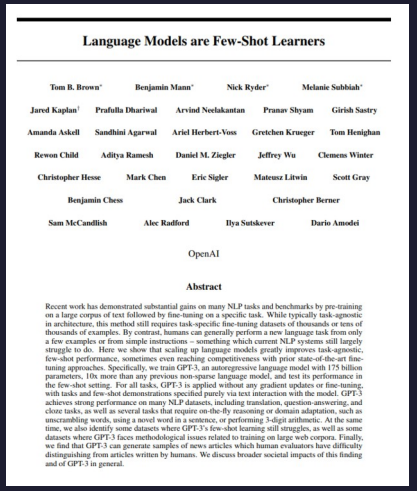
The dominant approach to creating ML systems is to collect a dataset of training examples demonstrating correct behavior for a desired task, train a system to imitate these behaviors, and then test its performance on independent and identically distributed (IID) held-out examples. This has served well to make progress on narrow experts. But the often erratic behavior of captioning models (Lake et al., 2017), reading comprehension systems (Jia & Liang, 2017), and image classifiers (Alcorn et al., 2018) on the diversity and variety of possible inputs highlights some of the shortcomings of this approach.

Our suspicion is that the prevalence of single task training on single domain datasets is a major contributor to the lack of generalization observed in current systems. Progress towards robust systems with current architectures is likely to require training and measuring performance on a wide range of domains and tasks. Recently, several benchmarks have been proposed such as GLUE (Wang et al., 2018) and decaNLP (McCann et al., 2018) to begin studying this.

Multitask learning (Caruana, 1997) is a promising framework for improving general performance. However, multitask training in NLP is still nascent. Recent work reports modest performance improvements (Yogatama et al., 2019) and the two most ambitious efforts to date have

2019 · GPT-2 · 1.5B

WebText 40GB · tareas sin entrenar



Language Models are Few-Shot Learners

Tom B. Brown* Benjamin Mann* Nick Ryder* Melanie Subbiah*
Jared Kaplan† Prafulla Dhariwal Arvind Neelakantan Pranav Shyam Girish Sastry
Amanda Askell Sandhini Agarwal Ariel Herbert-Voss Gretchen Krueger Tom Henighan
Rewon Child Aditya Ramesh Daniel M. Ziegler Jeffrey Wu Clemens Winter
Christopher Hesse Mark Chen Eric Sigler Mateusz Litwin Scott Gray
Benjamin Chess Jack Clark Christopher Berner
Sam McCandlish Alec Radford Ilya Sutskever Dario Amodei

OpenAI

Abstract

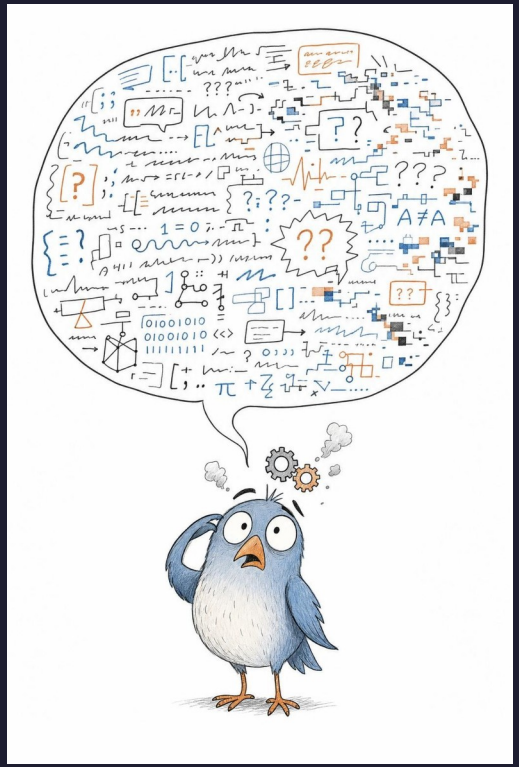
Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions – something which current NLP systems still largely struggle to do. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks, as well as several tasks that require on-the-fly reasoning or domain adaptation, such as unscrambling words, using a novel word in a sentence, or performing 3-digit arithmetic. At the same time, we also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora. Finally, we find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from articles written by humans. We discuss broader societal impacts of this finding and of GPT-3 in general.

2020 · GPT-3 · 175B

570GB · prompt engineering

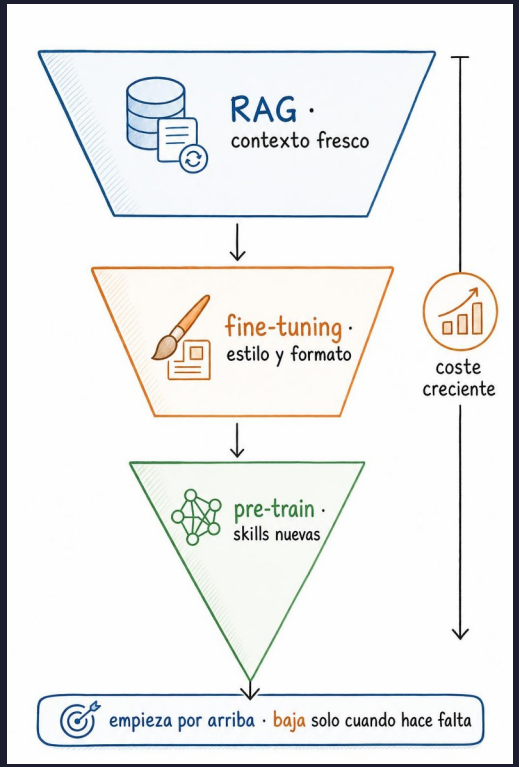
GPT-3 era potente... y descontrolado

- Prompt: 'Why aren't birds real?'
- GPT-3: 'The Rapture is not a very realistic event...'
- 'Let's say I was a bird. I've been a bird for a while now.'
- 'I have never flown... I am a bird in a cage.'
- Potente, pero delira y se contradice · le falta ALINEARSE (Deck 3)



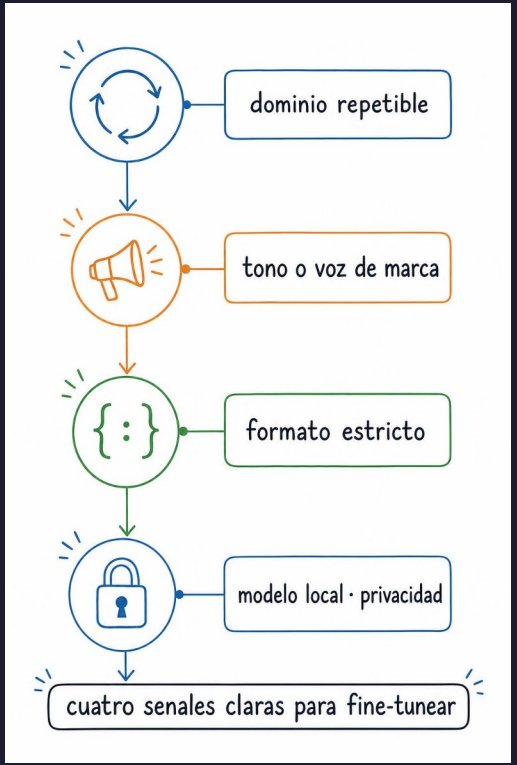
La pirámide de adaptación

- RAG · le das contexto fresco en cada query
- Fine-tuning · cambias estilo, tono y formato
- Pre-train · enseñas skills nuevas (carísimo)
- Empieza arriba · baja solo cuando hace falta



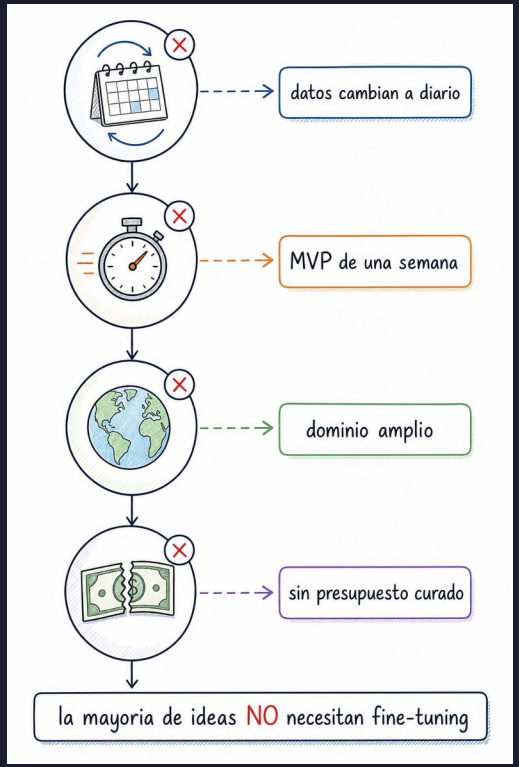
Cuándo SÍ fine-tunear

- Dominio repetible · NL->SQL, soporte, legal
- Tono o voz de marca · siempre idéntico
- Formato estricto · JSON con schema fijo
- Modelo local · privacidad o coste alto en API



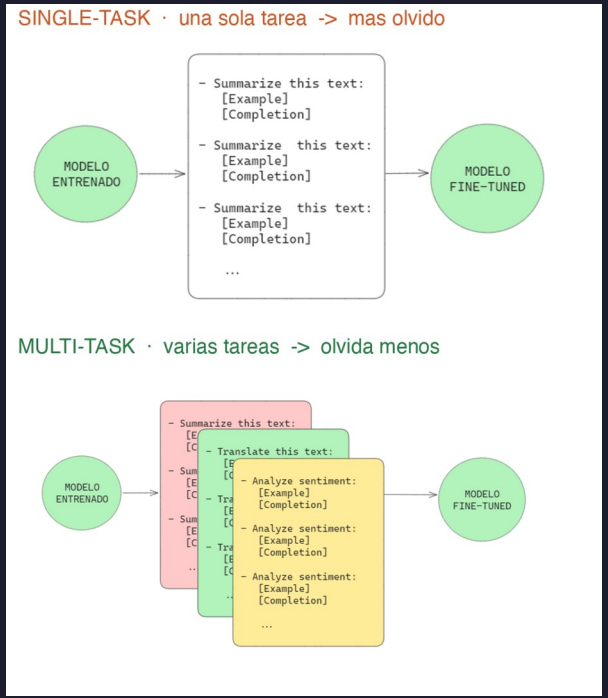
Cuándo NO fine-tunear

- Datos cambian cada día · mejor RAG
- MVP de una semana · prompt + few-shot
- Dominio amplio · ningún FT lo cubre
- La mayoría de ideas NO necesitan FT



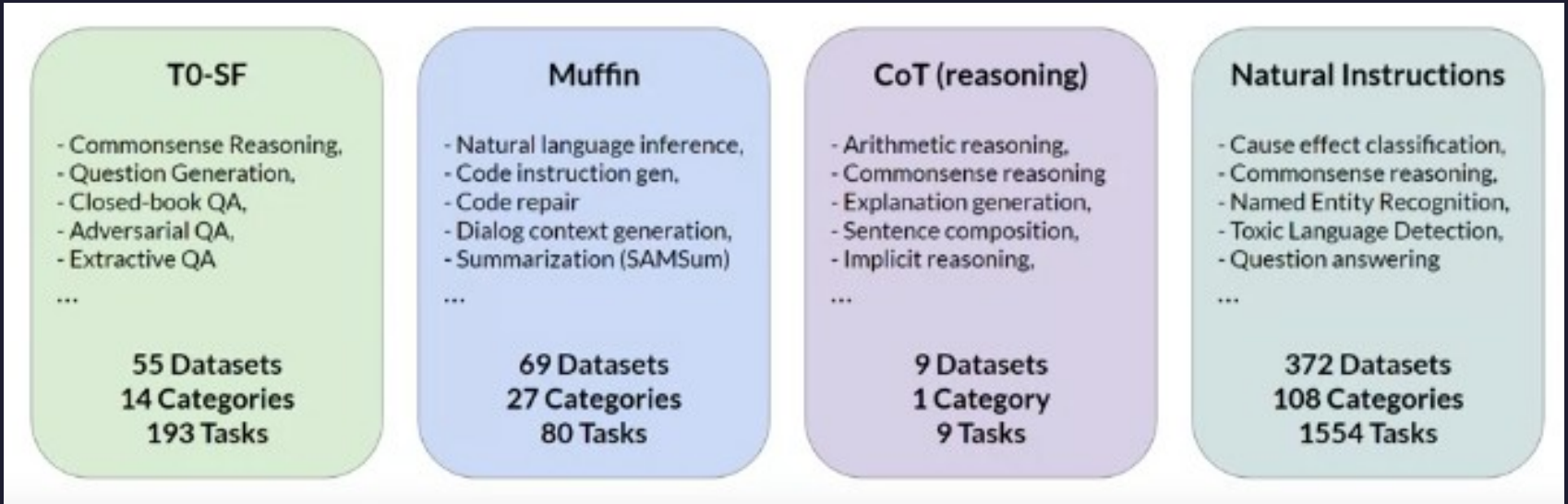
El riesgo: olvido catastrófico

- Al fine-tunear una tarea, el modelo mejora en ella
- ...pero puede empeorar en lo que ya hacía bien
- Mezcla un dataset general para mitigarlo
- El multi-task olvida menos que el single-task



Instruction tuning y FLAN

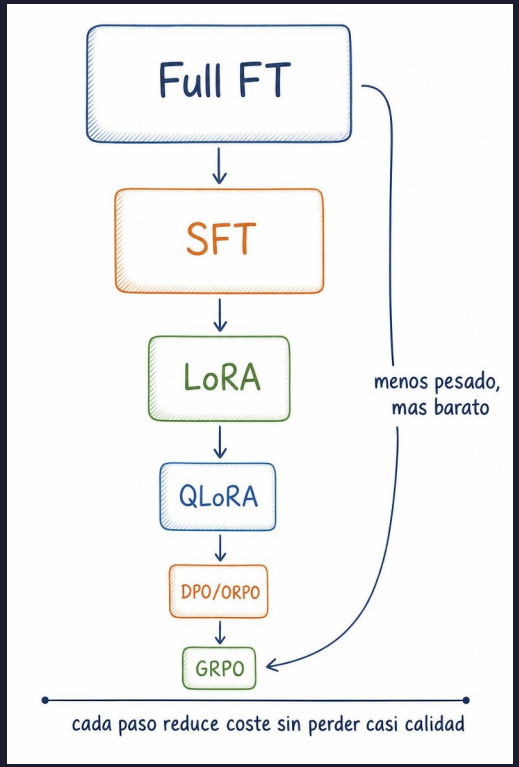
Single-task (precisión) vs multi-task (versátil, olvida menos). FLAN-T5: cientos de datasets en muchas categorías.



El cascade del fine-tuning

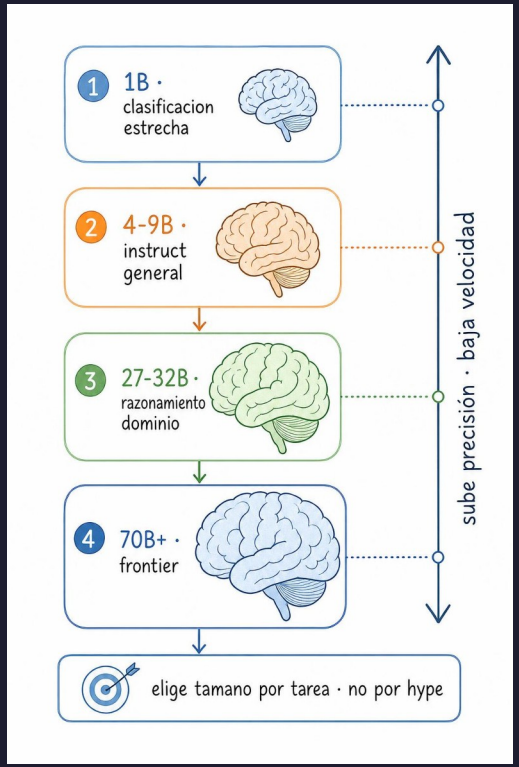
keep coding

- Full FT · todos los pesos
- SFT · supervisado con dataset etiquetado
- LoRA · transparencia sobre el modelo
- QLoRA · transparencia + cuantización 4-bit
- DPO/ORPO · alignment · GRPO · razonamiento



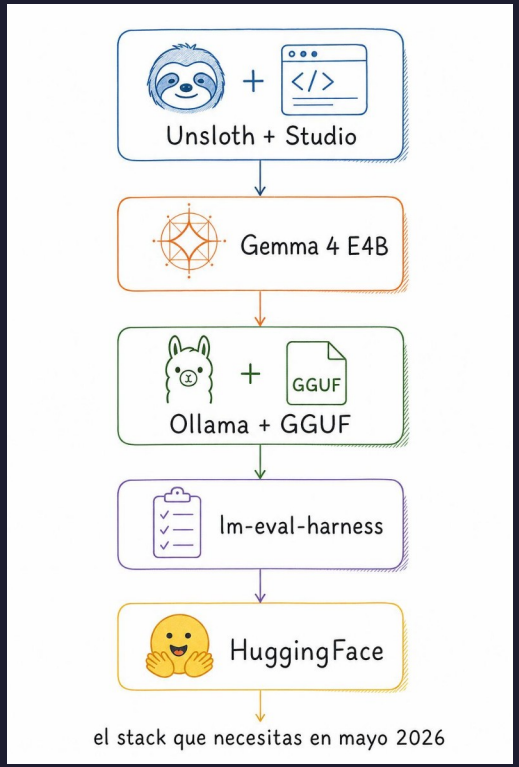
Mapeo modelo <-> tarea

- 1B (Gemma 3 1B) · clasificación estrecha
- 4-9B (Gemma 4 E4B, Qwen3.5-9B) · instruct general
- 27-32B (Qwen3.5-27B) · razonamiento de dominio
- 70B+ (Llama 4 Maverick) · frontier de dominio
- Sube si importa la precisión, baja si importa el coste



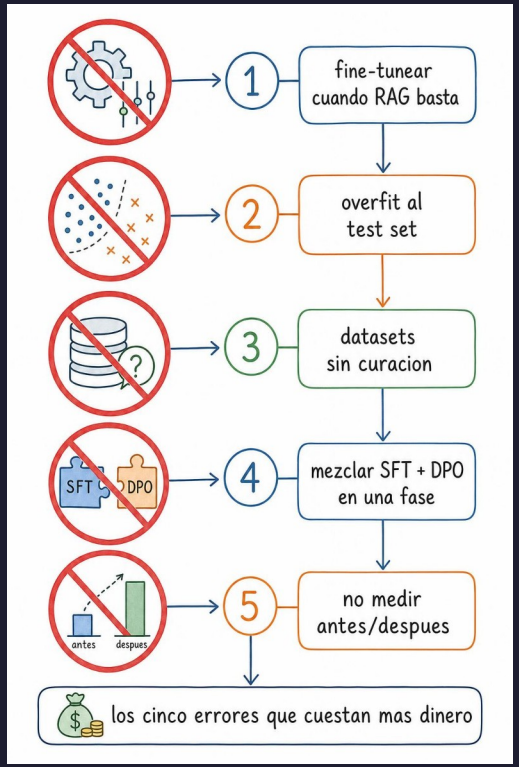
Stack 2026

- Unsloth · 2x más rápido, 70% menos VRAM
- Unsloth Studio · UI no-code (marzo 2026)
- Gemma 4 E4B · 128K contexto, multimodal (abril 2026)
- Ollama + GGUF para servir en local
- lm-evaluation-harness para medir después



Anti-patrones

- Fine-tunear cuando RAG sobra
- Overfit a tu propio test set
- Dataset sucio sin deduplicar ni filtrar
- Mezclar SFT y DPO en una sola fase
- No medir antes y después




keep coding